Lab 9: Implementation of the Codes for Developing GUI's in Java

Objective of the Lab:

To introduce students to **Graphical User Interface (GUI)** development using **Java Swing**, helping them understand how to create simple interactive windows with buttons, text fields, and other components.

What is GUI (Graphical User Interface)?

GUI stands for **Graphical User Interface**, which allows users to interact with programs through **visual elements** such as:

- Windows
- Buttons
- Text fields
- Checkboxes
- Dialog boxes

Instead of typing commands in (CLI), GUI makes programs **user-friendly** and easier to use

Basic Java Swing Components You Must Know

Component	Description
1. **JFrame**	Main window or screen of the app
2. **JLabel**	Displays text (non-editable)
3. **JTextField**	Allows user to type input (one line)
4. **JPasswordField**	Input for passwords (hidden text)
5. **JButton**	A button user can click to perform action
6. **JTextArea**	Input box for multiple lines (like feedback form)
7. **JCheckBox**	Multiple options selection (can select more than one)
8. **JRadioButton**	Single option selection among many
9. **JComboBox**	Dropdown menu (select one item)
10. **JPanel**	Grouping layout box helps organize GUI into sections
11. **JOptionPane**	Pops up small dialog box to show message or ask for input

How GUI Programs Work in Java (Step-by-Step)

1. **Import** Swing package using **import javax.swing.*;**
2. **Create a class** that extends **JFrame** or uses **JFrame** object.
3. **Add Components** like buttons, text fields, labels, etc.
4. **Add Action Listeners** to handle events like button clicks.
5. **Set Frame Properties** like size, title, visibility.

Component 1: JFrame Main Window

Purpose:

A JFrame is the **main window** where all other components (buttons, text fields, etc.) are added.

Real-Life Example:

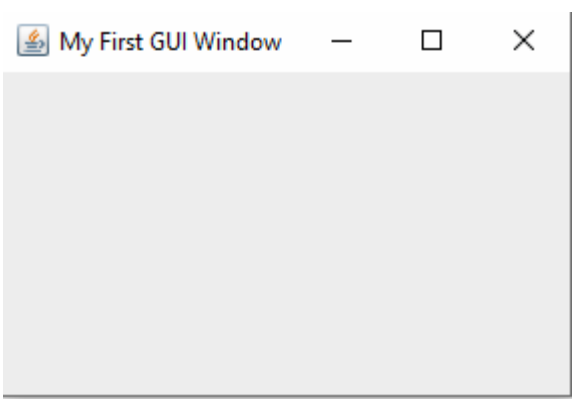
Jaise kisi food delivery app ka main screen hota hai usi tarah yeh main frame hota hai jisme sari cheezein show hoti hain.

Code:

```
import javax.swing.*;

public class FrameExample {
    public static void main(String[] args) {
        // JFrame ka object banaya jo main window represent karta hai
        JFrame frame = new JFrame("My First GUI Window");  // Window ka title diya
        frame.setSize(300, 200);  // Window ka size set kia (width x height)
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);  // Jab X button press ho to app band ho jaye
        frame.setVisible(true);  // Frame ko visible kia screen par
    }
}
```

Output:



Component 2: JLabel Static Text Display

Purpose:

A **JLabel** shows **text** that the user cannot edit. Useful for titles or instructions.

Real-Life Example:

Jaise kisi form mein "**Enter your name**" likha hota hai wo label hota hai.

Code:

```
import javax.swing.*;

public class LabelExample {
    public static void main(String[] args) {
        JFrame frame = new JFrame("Label Example");
        frame.setSize(300, 200);
        frame.setLayout(null);  // Layout manually set kar rahe hain
        // Label banaya aur usay position par set kia
        JLabel label = new JLabel("Hello, this is a label.");
        label.setBounds(50, 50, 200, 30);//(x, y, width, height)
        frame.add(label);  // Frame mein label add kia
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);  // Frame ko show kia
    }
}
```

Output:

Component 3: JTextField User Text Input

Purpose:

A **JTextField** allows the user to **type input** (like name or email).

Real-Life Example:

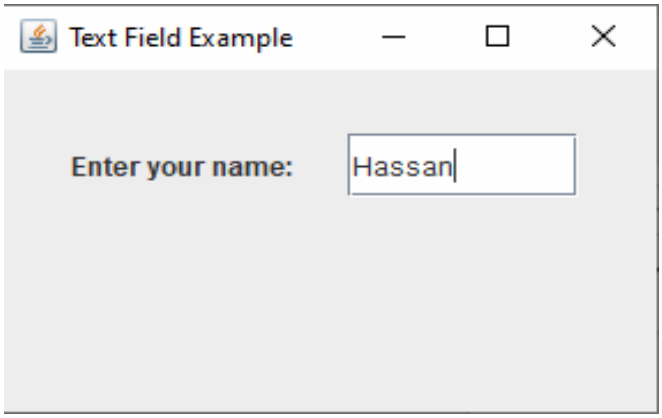
Login form mein jahan user apna naam ya email likhta hai wo text field hoti hai.

Code:

```
import javax.swing.*;

public class TextFieldExample {
    public static void main(String[] args) {
        JFrame frame = new JFrame("Text Field Example");
        frame.setSize(300, 200);
        frame.setLayout(null);
        // Label banaya jo user ko guide karega kya input karna hai
        JLabel label = new JLabel("Enter your name:");
        label.setBounds(30, 30, 120, 30);
        frame.add(label);
        // Text field jisme user text enter karega
        JTextField textField = new JTextField();
        textField.setBounds(150, 30, 100, 30);//(x, y, width, height)
        frame.add(textField); // Frame mein label add kia
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}
```

Output:



Component 4: JButton Clickable Button

Purpose:

A **JButton** performs an action when clicked (like Submit or Login).

Real-Life Example:

Kisi bhi online form mein "**Submit**" ya "**Next**" ka button hota hai wo JButton hota hai.

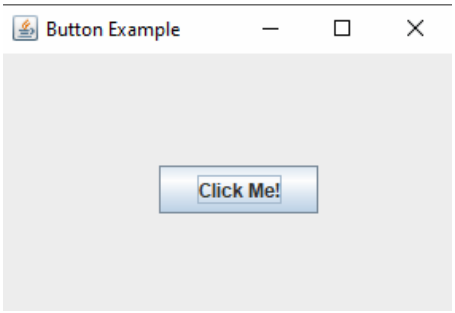
Code:

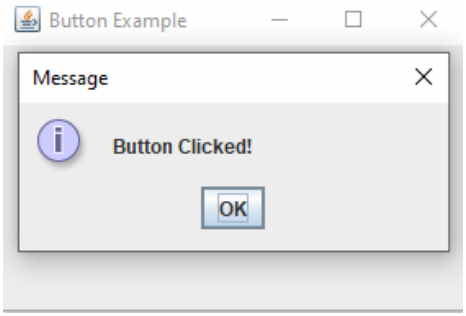
```
import javax.swing.*;
import java.awt.event.*;

public class ButtonExample {
    public static void main(String[] args) {
        JFrame frame = new JFrame("Button Example");
        frame.setSize(300, 200);
        frame.setLayout(null);
        // Button banaya
        JButton button = new JButton("Click Me!");
        button.setBounds(100, 70, 100, 30);
        frame.add(button);
        // Jab button click ho to message show ho
        button.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                JOptionPane.showMessageDialog(frame, "Button Clicked!");
            }
        });
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
```

```
    }
}
```

Output:


Button Example
Click Me!


Button Example
Message
Button Clicked!
OK

Component 5: JPasswordField Hidden Password Input

Purpose:

A **JPasswordField** lets the user enter a password that is shown as dots.

Real Life Example:

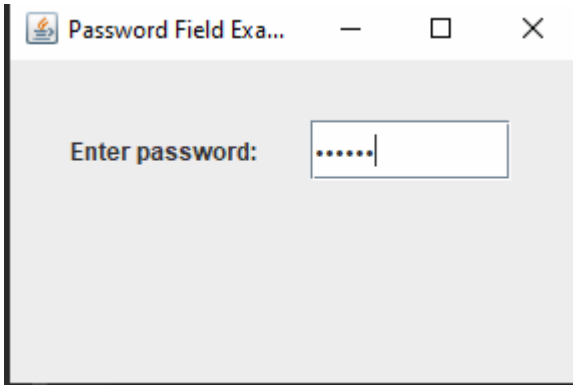
Login screen mein jab password likhte hain to dots ya stars show hoti hain wo password field hoti hai.

Code :

```
import javax.swing.*;

public class PasswordFieldExample {
    public static void main(String[] args) {
        JFrame frame = new JFrame("Password Field Example");
        frame.setSize(300, 200);
        frame.setLayout(null);
        // Password label
        JLabel label = new JLabel("Enter password:");
        label.setBounds(30, 30, 120, 30);
        frame.add(label);
        // Password field banaya jisme input hidden hoga
        JPasswordField passField = new JPasswordField();
        passField.setBounds(150, 30, 100, 30);
        passField.setEchoChar('#');
        frame.add(passField);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}
```

Output:



Component 6: JOptionPane Popup Box

Purpose:

JOptionPane is used to show a **popup message box** or take quick input.

Real-Life Example:

Jab app install karte waqt "Installation Successful" ka popup aata hai wo **JOptionPane** se banta hai.

Code

```
import javax.swing.*;

public class OptionPaneExample {
    public static void main(String[] args) {
        // Simple popup show karega message ke sath
        JOptionPane.showMessageDialog(null, "Welcome to GUI!");
    }
}
```

Output: